

Spirographs and Groups

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1. Introduction

Circles have perplexed mathematicians since the dawn of time. A shape defined with no vertices yet rudimentrally approximated by a polygon with $n \rightarrow \infty$ vertices. Any repeating pattern observed around us can either be modelled by a circle or a series of such (Fourier).

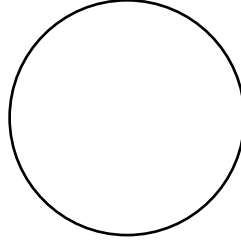


Figure 1: A circle

If such a simple object has mathematicians running around in circles for centuries, what should happen if two circles interacted with each other? Specifically, if a circle rolled, without slip around another circle, and path of a point on the circle is traced, what does it look like? How is it related to group theory? When a smaller circle rolls without slip inside another circle and a path is traced along some fixed point on the smaller circle, a hypotrochoid is traced. While spirographs are not always hypotrochoids, they are commonly synonymous as this paper treat them. In this paper, the concept of hypotrochoids and their relationship with group theory is explored.

2. Hypotrochoids

2.1. Derivation

In this paper, we are interested in the circles rolling, with no slip, around each other in Euclidian $\mathbb{R}^2$.

Supposition 1:

Define a revolution of a circle as the complete 2π rotation of the circle with respect to the x axis.

Suppose the following:

$\rho, r, R, \omega_r, t \in \mathbb{R} \quad 0 < r < R \text{ and } 0 < \rho$

r is the radius of the smaller r -circle

R is the radius of the bigger stationary R -circle

ρ is the radius of some fixed vector v from the center of r -circle

$\omega_r = 2\pi$ is the angular frequency of r -circle

t_r is number of revolutions r -circle has undergone from the initial conditions

t_R is the number of revolutions the center of r -circle has undergone from the initial conditions

In physics, angular frequency is defined as the number of revolutions per unit time. In this paper, it is defined as radians per revolutions so that the variable t can represent different units of interest, here revolutions.

Definition 1: Hypotrochoid

A hypotrochoid is the path traced by any fixed vector v relative to a circle of radius r rolling inside a circle of radius R . The vector v is at a fixed radius ρ from the center of the inner circle. A hypotrochoid is closed when there exists a number of revolutions such that the vector returns to its initial position.

Before thinking about a circle rolling inside of a larger circle. Let's consider the r -circle rolling on a flat line, the x -axis.

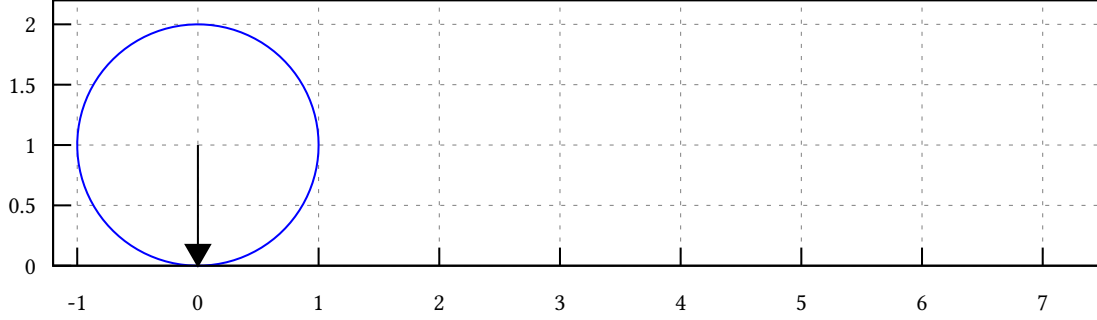


Figure 2: r -circle resting on the x -axis

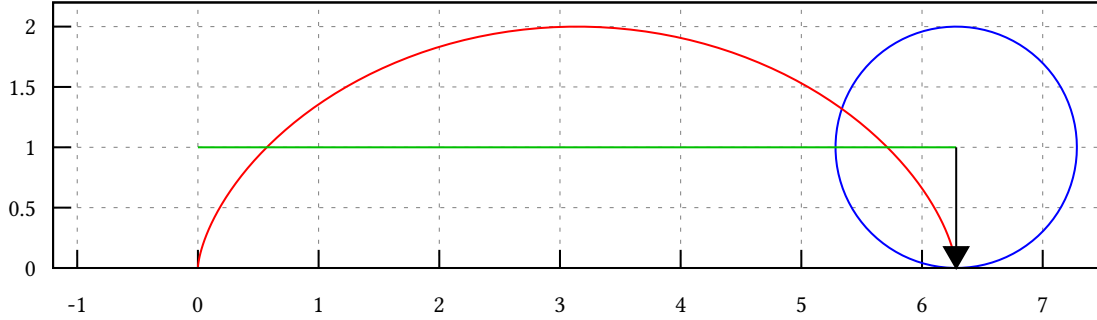


Figure 3: r -circle that rolled a whole revolution towards the right

Observe from Figure 3, the path traced by the vector v , with $\rho = r$ as illustrated by the vector and the center of the circle. Observe that in $t_r = 1$ revolutions, the arclength of the path traced by the center of the circle is exactly $2\pi r t_r$. In fact, we assume it the following:

Postulate 1:

In t rolls, t revolutions, of r -circle on any continuous line, the arclength of the path traced by the center of r -circle is $2\pi r t_r = r \omega t_r$.

Since we are also interested in the path traced by the vector v , we know that the translation of the vector is linear, that is a sum of its parts. The path traced by the vector v comes from both the rotation of the circle and the translation of the circle.

Note 1:

The parameterization of a vector v when a circle rotates clockwise around its center is:

$$\begin{aligned}x_r(t_r) &= \rho \cos(-\omega t_r) \\y_r(t_r) &= \rho \sin(-\omega t_r)\end{aligned}$$

Then, by linearity of translation and rotation, we have that the vector v is described by the following parameterization.

$$\begin{aligned}x_p(t_r) &= \rho \cos(-\omega t_r) + r\pi\omega t_r \\y_p(t_r) &= \rho \sin(-\omega t_r) + r\end{aligned}$$

Now, with what we learnt from r -circle rolling on the x -axis. Consider r -circle rolling on the inside of R -circle. Since, r -circle is always in contact with R -circle, the center of r -circle is always $R - r$ from the center of R -circle.

Note 2:

If r -circle rolls inside R -circle t revolutions, then the arclength covered by the center of r -circle is $2\pi(R - r)t_r = (R - r)\omega t_r$

To find out how many revolutions r -circle had to do in R -circle for every revolution of the center of r -circle, we divide the arclength of the path of the center of r -circle rolling inside R -circle by the arclength of the path of the center of r -circle in one revolution.

$$\begin{aligned}t_r &= \frac{2\pi(R - r)t_R}{2\pi r \cdot 1} && [\text{by Postulate 1 and Note 2}] \\&= \frac{R - r}{r}t_R && [\text{Algebra}]\end{aligned}$$

Since r -circle is rolling clockwise, its center therefore rotate counterclockwise with respect to the center of R -circle. The center of r -circle has therefore the following parameterization:

$$\begin{aligned}x_R(t_R) &= (R - r) \cos(\omega t_R) \\y_R(t_R) &= (R - r) \sin(\omega t_R)\end{aligned}$$

By linearity of translation and rotation, we get the parameterization of vector v .

$$\begin{aligned}x_p(t_R) &= x_r(t_r) + x_R(t_R) = x_r\left(\frac{R - r}{r}t_R\right) + x_R(t_R) \\y_p(t_R) &= y_r(t_r) + y_R(t_R) = y_r\left(\frac{R - r}{r}t_R\right) + y_R(t_R) \\&\Rightarrow x_p(t_R) = \rho \cos\left(-\omega \frac{R - r}{r}t_R\right) + (R - r) \cos(\omega t_R) \\&\text{and } y_p(t_R) = \rho \sin\left(-\omega \frac{R - r}{r}t_R\right) + (R - r) \sin(\omega t_R)\end{aligned}$$

Let $t = \omega t_R$, we have the parametric equations expressed in terms of the radians between the center of r -circle and the x -axis. Then,

$$\begin{aligned}
x_p(t) &= \rho \cos\left(-\frac{R-r}{r}t\right) + (R-r) \cos(t) \\
\text{and } y_p(t) &= \rho \sin\left(-\frac{R-r}{r}t\right) + (R-r) \sin(t) \\
\Rightarrow x_p(t) &= (R-r) \cos(t) + \rho \cos\left(\frac{R-r}{r}t\right) \quad [\cos(x) \text{ is even}] \\
\text{and } y_p(t) &= (R-r) \sin(t) - \rho \sin\left(\frac{R-r}{r}t\right) \quad [\sin(x) \text{ is odd}]
\end{aligned} \tag{1}$$

Definition 2: Lobe

For any hypotrochoid, a lobe of the hypotrochoid is a sector of the hypotrochoid that is repeated throughout the hypotrochoid. That is, any two lobes of a hypotrochoid is rotationally symmetric about the origin.

Theorem 1:

For any hypotrochoid, it is closed if and only if the fraction R/r is rational. Also, a closed hypotrochoid is bijective to a hypotrochoid with $R', r' \in \mathbb{Z}^+$ such that

$$\gcd(R', r') = 1 \text{ and } \frac{R'}{r'} = \frac{R}{r}$$

Proof: Suppose a hypotrochoid with ρ, r, R and the following parameterization

$$\begin{aligned}
x(t) &= (R-r) \cos(t) + \rho \cos\left(\frac{R-r}{r}t\right), \quad y(t) = (R-r) \sin(t) - \rho \sin\left(\frac{R-r}{r}t\right) \\
f(t) &= (x(t), y(t))
\end{aligned}$$

We know that cosine and sine repeats every 2π then for the vector v to return to its original position:

$$\text{Assume } t \in \mathbb{R}, \quad n \in \mathbb{Z}, \quad t = 2\pi n$$

Note that if $t = 0$, it is in its original position. So we consider when it is not.

$$\begin{aligned}
t &\equiv \frac{R-r}{r}t \pmod{2\pi} \\
\Leftrightarrow rt &\equiv (R-r)t \pmod{2\pi r} && [\text{mod properties}] \\
\Leftrightarrow rt &\equiv Rt \pmod{2\pi r} && [\exists k \in \mathbb{Z} \quad rt = 2\pi rk \text{ and } n = k \in \mathbb{Z}] \\
\Leftrightarrow rt &= Rt + 2\pi rm && [\text{mod defn; } m \in \mathbb{Z}] \\
\Leftrightarrow t &= \frac{R}{r}t + 2\pi m && [\text{Algebra}] \\
\Leftrightarrow t' &= \frac{R}{r}t && [t' = t - 2\pi m = 2\pi(n - m)] \\
\Leftrightarrow \frac{t'}{t} &= \frac{2\pi(n-m)}{2\pi n} = \frac{n-m}{n} = \frac{R}{r} \in \mathbb{Q} && [n, m, n-m \in \mathbb{Z}; \mathbb{Q} \text{ defn}] \\
\Leftrightarrow \exists a, b \in \mathbb{Z}^+ &\quad \gcd(a, b) = 1 \text{ and } \frac{a}{b} = \frac{R}{r}
\end{aligned}$$

The existence of a, b is guaranteed by assuming

$$a = \frac{R}{\gcd(R, r)}, \quad b = \frac{r}{\gcd(R, r)}$$

Suppose another hypotrochoid with ρ', r', R' where

$$R' = a, \quad r' = b \quad \text{and} \quad \rho' = \frac{R' - r'}{R - r} \rho$$

and thus its parameterization

$$\begin{aligned} x'(t) &= (R' - r') \cos(t) + \frac{R' - r'}{R - r} \rho \cos\left(\frac{R' - r'}{r'} t\right) \\ \text{and } y'(t) &= (R' - r') \sin(t) - \frac{R' - r'}{R - r} \rho \sin\left(\frac{R' - r'}{r'} t\right) \\ \text{and } g(t) &= (x'(t), y'(t)) \end{aligned}$$

Suppose the mapping

$$\begin{aligned} \psi : \text{Im}(f) &\mapsto \text{Im}(g) & \psi(x, y) &= \left(\frac{R' - r'}{R - r} x, \frac{R' - r'}{R - r} y \right) \\ &\because \forall (x, y) \in \text{Im}(f) \quad \exists t \in \mathbb{R} \quad f(t) = (x, y) \\ \Rightarrow \left(\frac{R' - r'}{R - r} x(t), \frac{R' - r'}{R - r} y(t) \right) &= (x'(t), y'(t)) & [x'(t), y'(t) \text{ definition}] \\ &= g(t) \in \text{Im}(g) & [g(t), \text{Im}(g) \text{ definition}] \end{aligned}$$

To show ψ is injective

$$\begin{aligned} &\text{Assume } (x_1, y_1), (x_2, y_2) \in \text{Im}(f) \quad (x_1, y_1) \neq (x_2, y_2) \\ &\Leftrightarrow x_1 \neq x_2 \text{ or } y_1 \neq y_2 \\ &\Rightarrow \frac{R' - r'}{R - r} x_1 \neq \frac{R' - r'}{R - r} x_2 \text{ or } \frac{R' - r'}{R - r} y_1 \neq \frac{R' - r'}{R - r} y_2 \\ &\quad [R' > r' \text{ and } R > r] \\ &\Rightarrow \left(\frac{R' - r'}{R - r} x_1, \frac{R' - r'}{R - r} y_1 \right) \neq \left(\frac{R' - r'}{R - r} x_2, \frac{R' - r'}{R - r} y_2 \right) \end{aligned}$$

To show ψ is surjective

$$\begin{aligned} &\text{Assume } (x', y') \in \text{Im}(g) \quad \exists t \in \mathbb{R} \quad g(t) = (x', y') \\ &\quad f(t) = (x(t), y(t)) \in \text{Im}(f) \\ &\Rightarrow \psi(x(t), y(t)) = (x'(t), y'(t)) & [\psi \text{ definition}] \\ &\quad = g(t) & [g(t) \text{ definition}] \end{aligned}$$

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Theorem 2:

For any hypotrochoid, if it is closed then it is rotationally symmetric and the larger radius of its coprime bijection is exactly the number of lobes.

Proof: For any closed hypotrochoid with R, r, ρ . Because lobes are rotationally symmetric about R -circle, by Definition 2, then the points where vector v is pointing outwards from the origin, as in the initial state, must be repeated. Crucially, this angle is the same angle as the center of r -circle about the origin. As r -circle is rolling counter-clockwise, the angle of the vector is

$$\begin{aligned}
& 2\pi - \frac{R-r}{r}t \\
\Rightarrow t &= 2\pi - \frac{R-r}{r}t \\
\Rightarrow rt &= 2\pi r - (R-r)t \\
\Rightarrow 0 &= 2\pi r - Rt \\
\Rightarrow Rt &= 2\pi r
\end{aligned}$$

By Theorem 1, this hypotrochoid is bijective to some hypotrochoid with R', r', ρ' such that

$$\begin{aligned}
\frac{R'}{r'} &= \frac{R}{r} \text{ and } \gcd(R', r') = 1 \\
\Rightarrow R't &= 2\pi r'
\end{aligned}$$

Since $R', r' \in \mathbb{Z}^+$ and $\gcd(R', r') = 1$, there are R' such lobes traced before the vector v returns to its initial state. Then it is symmetric. Notice the proof only concerns $R/r, R'/r'$, then all closed hypotrochoids and their coprime bijections have the same number of lobes.

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2.2. Examples

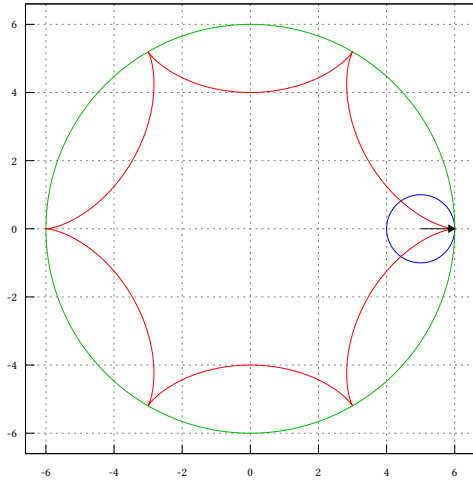


Figure 4: $t = 2\pi, R = 6, r = \rho = 1$

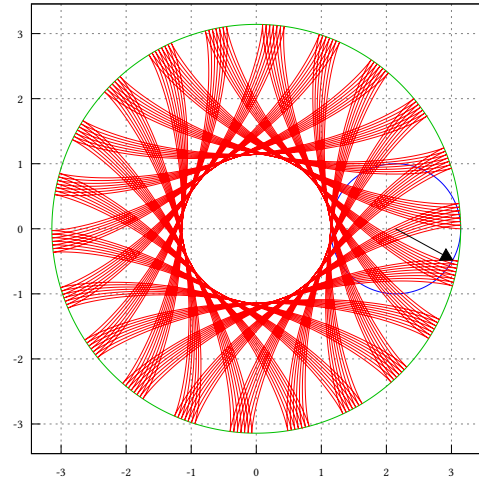


Figure 5: $t = 100\pi, R = \pi r = \rho = 1$

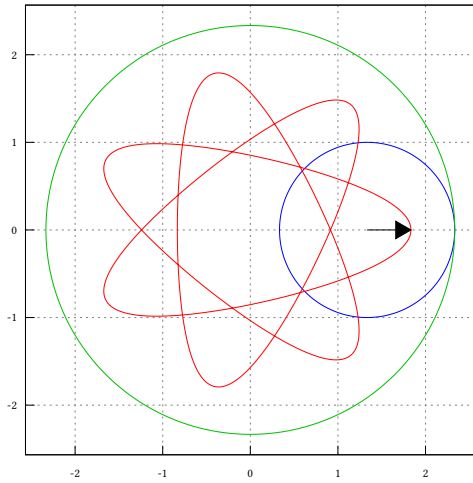


Figure 6: $t = 6\pi, R = \frac{7}{3}, r = 1, \rho = \frac{1}{2}$

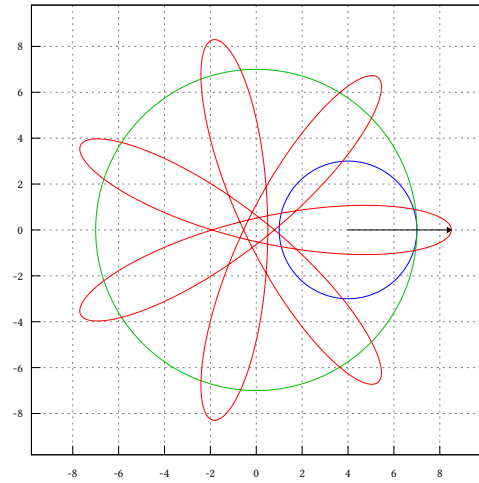


Figure 7: $t = 6\pi, R = 7, r = 3, \rho = \frac{7-3}{2(\frac{7}{3}-1)}$

Notice ψ from the proof of Theorem 1 applied to Figure 6 produces Figure 7. Counting the number of lobes in Figure 6, it is exactly R' as Theorem 2 predicts.

3. Group Theory

3.1. Isomorphism to Dihedral Groups

Observing Section 2.2, the hypotrochoids with rational R and r , seem to have some symmetry. Specifically rotational and reflectional symmetries. Suppose we only focus on the plane symmetries of each of the figures, we realise they are isomorphic to some D_n group.

Theorem 3:

For every closed hypotrochoid, its set of plane symmetries together with composition is isomorphic to a D_n group.

Proof: Given any closed hypotrochoid A . By Theorem 1, hypotrochoid A has a bijection to hypotrochoid A' with coprime radii. Suppose $R', r' \in \mathbb{Z}^+$ the respective coprime radii. By Theorem 2, number of lobes of the hypotrochoid A is the same of those hypotrochoid A' and the number of lobes is R' . Then, the hypotrochoid A has R' rotational symmetries. For any $a \in \mathbb{R}^+$, notice Equation 1 generates reflectionally symmetric path along $[-a, a]$ with respect to the x -axis. Then each lobe is reflectionally symmetric. Label the lobes $\{1, 2, \dots, R'\}$ consecutively in a counter-clockwise direction. Define

$$\tilde{r} = (1 \quad 2 \quad \dots \quad R') \text{ and } \tilde{s} = (2 \quad R')(3 \quad R' - 1) \dots$$

as bijections of $S_{R'}$. We have that

$$|\tilde{r}| = R' \text{ and } |\tilde{s}| = 2 \quad [\text{disjoint LCM theorem}]$$

Check that $\tilde{s}\tilde{r} = \tilde{r}^{R'-1}\tilde{s}$

$$\begin{aligned} \Rightarrow \tilde{s}\tilde{r} &= (1 \quad R')(2 \quad R' - 1)(3 \quad R' - 2) \dots \\ \because \tilde{r}^{R'-1} &= (1 \quad R' \quad R' - 1 \quad R' - 2 \quad \dots \quad 3 \quad 2) \\ \Rightarrow \tilde{r}^{R'-1}\tilde{s} &= (1 \quad R' \quad R' - 1 \quad \dots)(2 \quad R')(3 \quad R' - 1) \dots \\ &= (1 \quad R')(2 \quad R' - 1)(3 \quad R' - 2) \dots \\ \Rightarrow \tilde{s}\tilde{r} &= \tilde{r}^{R'-1}\tilde{s} \\ \text{Define } G &= \{ \langle \tilde{r}, \tilde{s} \rangle \mid |\tilde{r}| = R' \text{ and } |\tilde{s}| = 2 \text{ and } \tilde{s}\tilde{r} = \tilde{r}^{R'-1}\tilde{s} \} \\ &= \{ \tilde{e}, \tilde{r}, \tilde{r}^2, \dots, \tilde{r}^{R'-1}, \tilde{s}, \tilde{r}\tilde{s}, \tilde{r}^2\tilde{s}, \dots, \tilde{r}^{R'-1}\tilde{s} \} \end{aligned}$$

Note

$$\begin{aligned} D_{R'} &= \{ \langle r, s \rangle \mid |r| = R' \text{ and } |s| = 2 \text{ and } sr = r^{R'-1}s \} \\ &= \{ e, r, r^2, \dots, r^{R'-1}, s, rs, r^2s, \dots, r^{R'-1}s \} \end{aligned}$$

Suppose the mapping

$$\begin{aligned} \varphi : G &\mapsto D_{R'} \\ \tilde{a} &\rightarrow a \end{aligned}$$

To show it is a homomorphism

Assume $\tilde{a}, \tilde{b} \in G$

$$\varphi(\tilde{a}\tilde{b}) = ab \quad [\varphi \text{ definition}]$$

$$\begin{aligned}
&= (a)(b) && [\text{associativity of } * \text{ in } D_{R'}] \\
&= \varphi(\tilde{a})\varphi(\tilde{b}) && [\varphi \text{ definition}]
\end{aligned}$$

To show it is injective

$$\begin{aligned}
&\text{Assume } \tilde{a}, \tilde{b} \in G \quad \tilde{a} \neq \tilde{b} \\
&\Rightarrow \varphi(\tilde{a}) = a && [\varphi \text{ definition}] \\
&\quad \neq b \\
&\quad = \varphi(\tilde{b}) && [\varphi \text{ definition}]
\end{aligned}$$

To show it is surjective

$$\begin{aligned}
&\text{Assume } a \in D_{R'} \quad \text{Consider } \tilde{a} \\
&\Rightarrow \varphi(\tilde{a}) = a && [\varphi \text{ definition}] \\
&\quad \therefore G \cong D_{R'}
\end{aligned}$$

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Corollary 1:

For any closed hypotrochoid, its rotational symmetries is applicable under the Fundamental Theorem of Cyclic Groups.

Proof: By Theorem 3, we can see that its plane symmetries is isomorphic to a D_n group and its rotational symmetry has its cyclic subgroup of order $R' \in \mathbb{Z}^+$. Then, it satisfies the antecedent of the Fundamental Theorem of Cyclic Groups and therefore applicable under which.

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3.2. The Circle Group

In Supposition 1, the class of hypotrochoids have been restricted to non-degenerate only. If any of the suppositions are violated, degenerate hypotrochoids are formed. Specifically, suppose $\rho = 0$. Then, the path traced by the vector v of norm 0 is just the path traced by the center of the circle. Indeed, by Equation 1 if $\rho = 0$, the path traced is a circle.

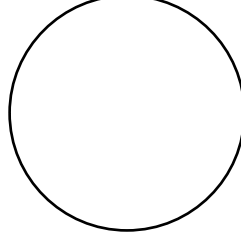


Figure 8: Another circle

Inspecting the circle closely, one might realise that there does not seem to have any lobes. Therefore, by deduction, its plane symmetries is not isomorphic to any dihedral group. Alternatively, one might also think there are an infinite number of axis of symmetry and therefore isomorphic to D_∞ . While it may be tempting to think so, the number of axis of symmetry is uncountable but D_∞ is defined on a countably infinite. So, the symmetries of the circle is not isomorphic to D_∞ .

Observe that the any rotation on the circle is symmetric. If one were to rotate it exactly 2π , any labelled points on the circle return to its initial state. If it were a group, it has the “cyclic” property. Yet, by definition of cyclic, the rotations must be discrete and countable, to which this is not.

Definition 3: Circle Group

The circle group is the set $\mathbb{T}$ with the $*$ operator as defined

$$\mathbb{T} = \{z \in \mathbb{C} \mid |z| = 1\}$$

$$* : \mathbb{T} \times \mathbb{T} \mapsto \mathbb{T}$$

$$e^{iz_1} * e^{iz_2} = e^{i(z_1+z_2)}$$

We will check that this is indeed a group

Note that $e^{i0} = 1 \in \mathbb{T}$ as $|1| = 1$ then $\mathbb{T}$ is non-empty.

To show closure

Assume $e^{i\theta_1}, e^{i\theta_2} \in \mathbb{T}$

$$\begin{aligned} \Rightarrow |e^{i\theta_1} e^{i\theta_2}| &= |e^{i(\theta_1+\theta_2)}| \\ &= \sqrt{\cos(\theta_1 + \theta_2)^2 + \sin(\theta_1 + \theta_2)^2} && \text{[By complex modulus definition]} \\ &= \sqrt{1} && \text{[Trig identity]} \\ &= 1 \\ \Rightarrow e^{i\theta_1}, e^{i\theta_2} &\in \mathbb{T} \end{aligned}$$

To show associativity

Assume $e^{i\theta_1}, e^{i\theta_2} e^{i\theta_3} \in \mathbb{T}$

$$\Rightarrow e^{i\theta_1} (e^{i\theta_2} e^{i\theta_3}) = e^{i\theta_1} e^{i(\theta_2+\theta_3)}$$

$$\begin{aligned}
&= e^{i(\theta_1+\theta_2+\theta_3)} \\
&= e^{i(\theta_1+\theta_2)} e^{i\theta_3} \\
&= (e^{i\theta_1} e^{i\theta_2}) e^{i\theta_3}
\end{aligned}$$

To show identity existence

$$\begin{aligned}
&\text{Assume } e^{i\theta} \in \mathbb{T} \quad \text{Consider } e^{i0} = 1 \in \mathbb{T} \\
&\Rightarrow e^{i\theta} e^{i0} = e^{i\theta} 1 = e^{i\theta} = 1 e^{i\theta} = e^{i0} e^{i\theta}
\end{aligned}$$

To show inverse existence

$$\begin{aligned}
&\text{Assume } e^{i\theta} \in \mathbb{T} \quad \text{Consider } e^{-i\theta} \\
&\because |e^{-i\theta}| = \sqrt{\cos(-\theta)^2 + \sin(-\theta)^2} = \sqrt{1} = 1 \\
&\Rightarrow e^{i-\theta} \in \mathbb{T} \text{ and } e^{i\theta} e^{-i\theta} = e^{i0}
\end{aligned}$$

Therefore $\mathbb{T}$ is a group by group axioms.

4. Conclusion

In this article, we restrict the class of spirographs in to only hypotrochoids. From which, we have proven the existence of bijections of any closed hypotrochoid to one made with coprime integer radii, Theorem 1. Which leads to the proof of the number of lobes any closed hypotrochoid is exactly the larger coprime radius of its bijection, Theorem 2. From these two theorems, it is proven the existence of an isomorphism between the plane symmetries of any closed hypotrochoid and some dihedral group, Theorem 3. Furthermore, the Fundamental Group of Cyclic Groups is discovered to be applicable to the rotational symmetries of a closed hypotrochoid, Corollary 1. In the case, when $\rho = 0$, the rotational symmetries of the degenerate hypotrochoid, the circle, is proven to be a group itself. We excluded many cases of spirographs such as other degenerate hypotrochoids and epitrochoids. Although on further examination, one should find many of the theorems proven applicable to those spirographs too. Perhaps, there is a “Fundamental Theorem of Spirographs” but that remains to be found.